

Vehicle Stability Control for Independent Wheel Drive Systems with optimal torque-allocation using MPC

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Abstract

This paper presents a proposed integrated vehicle stability control strategy for a four-wheel-independent-drive electric vehicle using a two-layer hierarchical control architecture consisting of an upper-layer fuzzy logic based direct yaw moment controller, a second-layer model predictive controller for optimal wheel torque distribution. In the proposed scheme, the upper layer generates the required corrective yaw moment based on the deviation between actual and desired yaw rates and slip angles, while the MPC layer determines the optimal torque correction values for the four wheels by solving a constrained quadratic programming problem in real time. The MPC is formulated using a prediction model based on wheel-slip dynamics and yaw-moment generation, with the wheel slip ratios as state variables, individual motor torques as control variables, and corrective yaw moment tracking as the main control objective. The optimization incorporates actuator and stability-related constraints, including motor torque limits, torque increment limits, vehicle yaw-moment bounds, and slip-ratio limits to preserve tire-road adhesion and prevent excessive wheel slip. The proposed method is intended to improve yaw-rate tracking, reduce sideslip, and maintain tire-road adhesion under demanding driving conditions. The controller is implemented in MATLAB Simulink and evaluated against a standard double lane change maneuver and sinusoidal dwell maneuver formulated in CarSim.

Introduction

Electric vehicles with independently actuated wheels offer significant potential for advanced vehicle dynamics control, as each wheel's torque can be adjusted rapidly and independently. This capability enables direct control of yaw motion, lateral stability, and traction, particularly under aggressive maneuvers or low-adhesion conditions. However, this added flexibility also increases system complexity, requiring effective coordination between wheel torques, tire forces, and overall vehicle motion. As a result, integrated stability control has become a key challenge in distributed-drive EVs [1].

Recent research has increasingly focused on hierarchical and model-based control approaches to manage this complexity and improve performance. Model predictive control (MPC) has emerged as a particularly effective method due to its ability to incorporate system dynamics and explicitly handle constraints such as motor torque limits and tire adhesion boundaries while optimizing control actions in real time [2]. At the same time, studies on torque distribution highlight that the lower-layer control allocation is not merely a secondary step but a decisive factor influencing yaw-rate tracking, stability margins, and overall vehicle response under varying conditions [3,4]. These insights motivate the development of integrated control architectures, where high-level controllers determine desired vehicle behavior and lower-level controllers allocate actuator efforts efficiently. Accordingly, this work proposes a hierarchical strategy that combines a fuzzy logic upper layer for yaw-moment generation with an MPC-based torque allocation layer, aiming to achieve robust and constraint-aware stability control in four-wheel-independent-drive electric vehicles.

I. System Model Framework

To investigate the lateral stability control problem of the four-wheel-independent-drive electric vehicle, an integrated control-oriented model was developed comprising a CarSim model which represents the plant vehicle dynamics model, a reference model for desired yaw rate, upper-level fuzzy controller and lower layer MPC for torque allocation.

A. Plant Vehicle Dynamics CarSim Model

In this work, the vehicle dynamics is represented by a high-fidelity CarSim D-Class SUV model, which serves as the main nonlinear plant in the control loop. The CarSim block receives the four commanded wheel torques:

$$\mathbf{u}(k) = [T_{fl}(k) \quad T_{fr}(k) \quad T_{rl}(k) \quad T_{rr}(k)]^T \quad (1)$$

and provides the main vehicle inputs required by the controller,

$$\mathbf{y}(k) = [\gamma(k) \quad \beta(k) \quad \delta(k) \quad V_x(k) \quad \omega_i(k) \quad T_i(k) \quad F_{zi}(k)]^T \quad (2)$$

where γ is the actual yaw rate, β is the vehicle sideslip angle, δ is the steering angle, ω_i is the wheel speed, S_i is the wheel slip ratio, F_{zi} is the vertical load at each wheel. V_x is the longitudinal speed, and T_i is the torque output at each wheel. This arrangement allows the controller to interact directly with a realistic vehicle model. The vehicle parameters are attached in Appendix-1.

B. Two Degrees of Freedom: Reference Model

A two-degree-of-freedom reference model is used to generate the desired handling states for the upper-layer stability controller. The desired yaw rate is defined as

$$\gamma_d = \frac{V_x}{L(1 + KV_x^2)} \delta \quad (3)$$

where V_x is the longitudinal velocity, L is the wheelbase, K is the stability factor, and δ is the front road-wheel angle. The desired body slip angle is selected as

$$\beta_d = 0 \quad (4)$$

which reflects the objective of maintaining minimal sideslip during stability-critical operation. The reference model defines the ideal lateral response that the controller seeks to track.

II. Control Framework

A. Upper-Level Fuzzy Controller

The upper-layer controller is formulated as a Mamdani-type fuzzy controller [1] with two inputs and one output. The output of the fuzzy controller is the additional corrective yaw moment $\mathbf{M}$ required to regulate the vehicle lateral motion:

$$\mathbf{M} = \mathcal{F}(\Delta\gamma, \Delta\beta) \quad (5)$$

where $\mathcal{F}(\cdot)$ represents the nonlinear fuzzy mapping between the state-tracking errors and the commanded direct yaw moment.

Side Slip Angle Error $\Delta\beta$	Yaw rate Error $\Delta\dot{y}$						
	NE	NM	NL	ZE	PL	PM	PE
NE	NE	NE	NM	NM	NL	PM	PE
NM	NE	NE	NM	NL	ZE	PM	PE
NL	NE	NM	NL	NL	ZE	PM	PE
ZE	NE	NL	NL	ZE	PL	PL	PE
PL	NE	NM	ZE	PL	PL	PM	PE
PM	NE	NM	ZE	PL	PM	PE	PE
PE	NE	NM	PL	PM	PM	PE	PE

Table 1: Fuzzy Control Logic

Prior to fuzzification, both input variables are subjected to saturation and normalization to ensure consistent operation of the fuzzy inference mechanism within a bounded domain. The yaw-rate error is first limited within the interval $[-10, 10]$, after which it is normalized to the range $[-1, 1]$. Similarly, the sideslip-angle error is constrained within $[-6, 6]$ and then scaled to the same normalized interval $[-1, 1]$. Both input variables are described using seven linguistic terms: Negative Extreme (NE), Negative Medium (NM), Negative Low (NL), Zero (ZE), Positive Low (PL), Positive Medium (PM), and Positive Extreme (PE). Trapezoidal membership functions are used for the inputs to provide smooth transitions between linguistic regions and to maintain sensitivity across a wide operating range. The output yaw moment is represented using Gaussian membership functions, which allow smoother and more precise control action during defuzzification. The fuzzy rule base defines the relationship between the input errors and the required corrective yaw moment. The logic of the rule table [Table 1] is constructed based on the principle of stabilizing the vehicle by counteracting deviations in yaw rate and sideslip.

B. Model Predictive Torque Allocator Controller

The lower-layer controller was formulated using model predictive control [1] to allocate the corrective yaw-moment demand among the four independently driven wheels. The role of this controller is to determine the wheel-torque correction sequence that best satisfies the required yaw-moment response while respecting actuator limits and wheel-slip constraints. The wheel slip ratio of each wheel is defined as:

$$S_i = \frac{\omega_i R - V_x}{V_x} \quad (6)$$

where S_i is the slip ratio of wheel i . Since wheel slip is closely related to the available longitudinal tire force and vehicle stability, it is selected as the state variable in the predictive control model. The system was then discretized where $\Delta x(k)$ is the state increment vector, $\Delta u(k)$ is the control increment vector, and $\Delta y(k)$ is the output increment vector. The control increment vector is composed of the torque corrections applied to the four wheels, i.e.,

$$\Delta u(k) = \begin{bmatrix} \Delta T_{fl}(k) \\ \Delta T_{fr}(k) \\ \Delta T_{rl}(k) \\ \Delta T_{rr}(k) \end{bmatrix} \quad (7)$$

where ΔT_{fl} , ΔT_{fr} , ΔT_{rl} , and ΔT_{rr} denote the torque increments of the front-left, front-right, rear-left, and rear-right wheels, respectively. Over the prediction horizon, the future output sequence can be expressed in compact form as

$$Y(k+1|k) = S_c \Delta x(k) + Z_c U(k) + I_y y(k) \quad (8)$$

where S_c and Z_c are the prediction matrices, $U(k)$ is the future control sequence over the control horizon, and $I_y y(k)$ represents the contribution of the current output to the predicted output sequence. The control objective is formulated as a quadratic cost function that penalizes both output-tracking error and excessive control effort. Accordingly, the MPC optimization problem is written as

$$J = \| P(Y(k+1|k) - Y_r(k+1|k)) \|^2 + \| E U(k) \|^2 \quad (9)$$

where $Y_r(k+1|k)$ is the reference output sequence associated with the required corrective yaw moment, P is the output-weighting matrix, and E is the control-weighting matrix. The first term enforces tracking of the desired yaw-moment behavior, while the second term suppresses excessive wheel-torque variation.

$$J = \frac{1}{2} U^T H U + G^T(k+1|k) U \quad (10)$$

To enable real-time implementation, the QP problem is solved using the Hildreth dual method [5], which is an iterative algorithm based on Lagrange multipliers. In this approach, the constrained QP problem is converted into its dual form, and the optimal solution is obtained through successive updates of the dual variables until convergence.

The MPC controller continuously reallocates wheel torques in real time according to the current vehicle state, the required corrective yaw moment, and the active system constraints.

Parameter	Symbol	Value
Sampling time	T_s	0.025 s
Prediction horizon	N_p	10
Control horizon	N_c	3
Output weight	W_y	1000.0
Control-effort weight	W_u	0.01
Max wheel torque	$T_{\max}$	[500 Nm, -500Nm]
Torque rate limit	$\Delta T_{\max}$	200 Nm/step
Slip-ratio bounds	[$S_{\min}$, $S_{\max}$]	[-0.2, 0.2]

Table 2: MPC tuning parameters used for all test cases

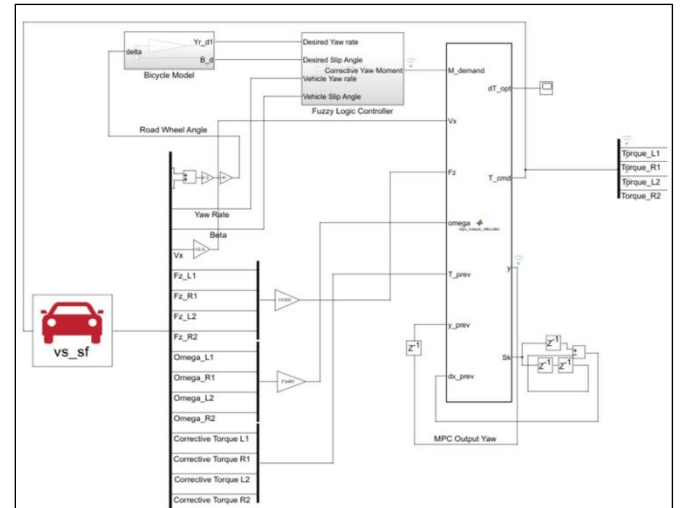


Figure 3: Simulink and Carsim integrated model

Simulation Analysis

The three test cases used in this evaluation were selected to exercise three distinct regimes of controller operation. The tight DLC at high adhesion stresses transient yaw response under rapid steering reversals while remaining within the friction envelope, the DLC at low adhesion brings the slip-ratio constraint into active use and exposes any model-mismatch issues in the MPC's linear tire-stiffness assumption and the sine wave maneuver evaluates the controller against an internationally recognized regulatory stability standard. All three are available as preset maneuvers in CarSim, with steering input generated by CarSim's internal driver model rather than a Simulink-side driver, ensuring repeatability across the controlled and uncontrolled comparison runs.

A. Test Case 1: Tight Double Lane Change Maneuver

The first test case is a tight ISO 3888-2 double-lane-change performed on a high-adhesion road surface ($\mu = 0.85$). The vehicle enters the maneuver at 120 km/h, with CarSim's internal driver model controlling steering input. The same maneuver is run twice: once with the controller active and once with the controller disabled (zero corrective torque demand on all four wheels) to obtain the uncontrolled baseline.

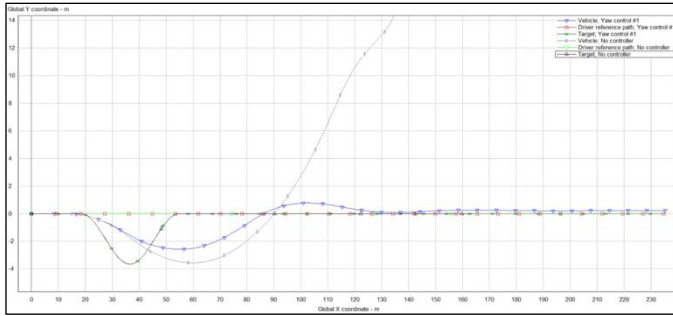


Figure 4: Reference Trajectory vs MPC tracking

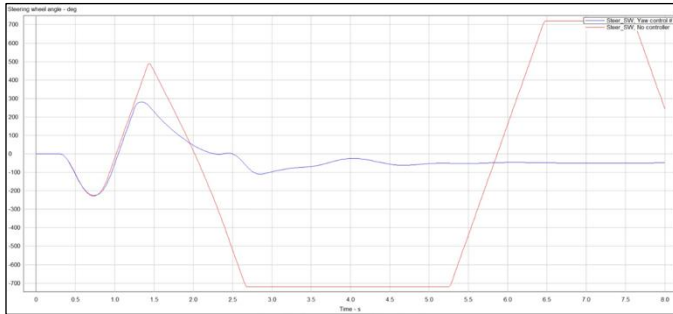


Figure 5: Steering Wheel Angle for Tight DLC maneuver

The vehicle trajectory [Fig 4] during the tight DLC maneuver shows that the proposed controller is able to follow the desired path with a lot better stability than the uncontrolled case. Compared to the uncontrolled case, the controlled vehicle exhibits significantly reduced path overshoot and improved path recovery after the lane transition. The steering input [Fig 5] profile indicates that both controlled and uncontrolled cases are subjected to the same driver steering command, ensuring a fair comparison. The steering angle comparison shows that the controller minimizes the steering effort to keep the vehicle stable.

Results

The yaw moment tracking plot [Fig 7] results show that the MPC controller effectively follows the desired yaw moment generated by the fuzzy controller. Although minor deviations are observed during rapid transients, the overall tracking performance is satisfactory. The controller is able to generate the required corrective yaw moment in

real time, ensuring that the vehicle response remains close to the desired behavior.

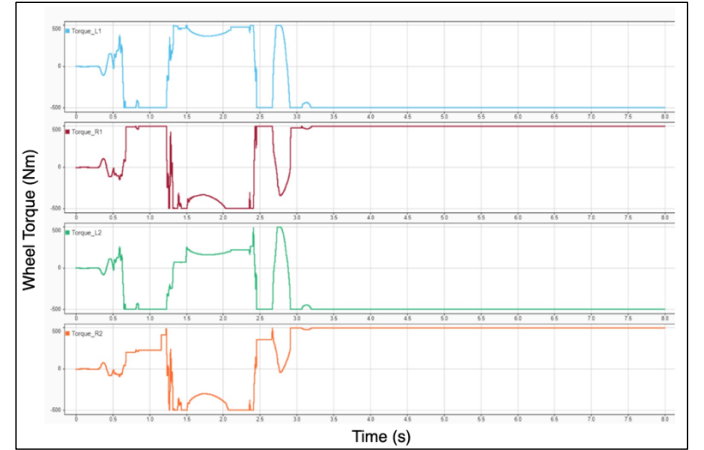


Figure 6: Wheel torque (1. Front left wheel 2. Front right wheel 3. Rear left wheel 4. Rear right wheel) vs time for tight DLC

The wheel torque profiles [Fig 6] illustrate the active redistribution of torque among the four wheels. During the maneuver, significant variations in individual wheel torques are observed, particularly during high dynamic phases. The controller applies differential torque between left and right wheels to generate the required yaw moment while maintaining stability. After the transient phase, the torques stabilize, indicating convergence of the control action. This demonstrates the effectiveness of the MPC-based torque allocation in coordinating longitudinal and lateral dynamics. The maximum torque allocation constraint is followed and the motor does not overshoot the 500 Nm upper limit.

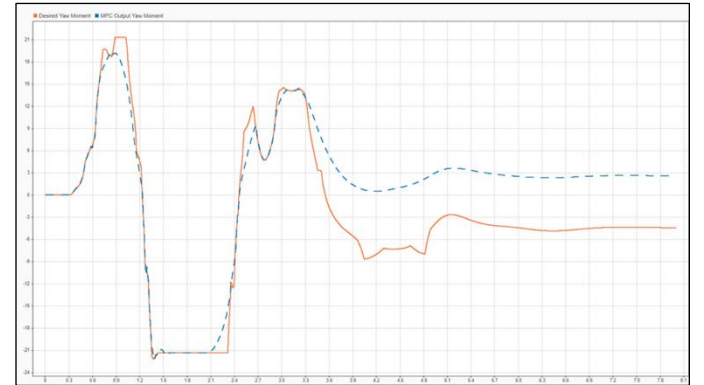


Figure 7: Yaw moment tracking for tight DLC

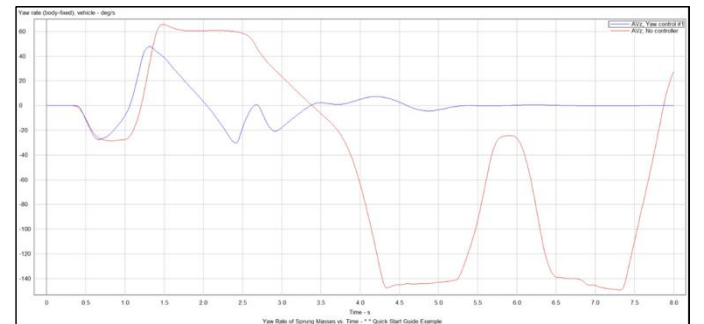


Figure 8: Yaw rate tracking (with MPC vs w/o controller)

The yaw rate response [Fig 8] demonstrates a clear improvement with the proposed MPC-based controller. Without control, the vehicle

exhibits large oscillations and overshoot, particularly during rapid lane transitions. With the controller, the yaw rate closely follows the desired response, with reduced peak values and faster settling time. This indicates improved yaw stability and better tracking of the reference model, consistent with the behavior observed in similar MPC-based strategies.

The sideslip angle [Fig 9] is significantly reduced under the proposed control strategy. In the uncontrolled case, large deviations in sideslip angle are observed, indicating potential instability and loss of lateral control. With the controller, the sideslip angle remains close to zero throughout the maneuver, demonstrating effective suppression of lateral deviation and improved vehicle stability. This confirms that the controller successfully regulates both yaw rate and sideslip simultaneously.

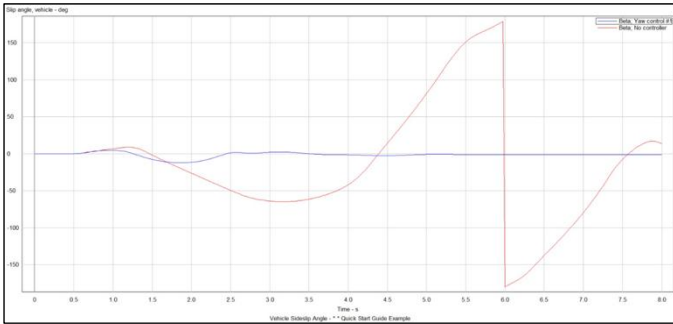


Figure 9: Side Slip Angle (with MPC vs w/o controller)

B. Test Case 2: Double Lane Change Maneuver at Low Adhesion Road Surface

The second test case uses the same DLC geometry but reduces the road adhesion coefficient to $\mu = 0.3$, simulating a wet or icy surface. The entry speed is reduced to 80 km/h to keep the maneuver feasible for the uncontrolled baseline, at higher entry speeds the uncontrolled vehicle departs the lane within the first turn-in. This case is the most demanding for the controller because the slip-ratio constraint becomes binding and the linear tire-stiffness assumption used in the MPC prediction model degrades. Reference path for the DLC maneuver is given in Fig 11.

The trajectory response clearly shows the benefit of the controller under this severe condition. With the proposed controller, the vehicle is able to complete the lane-change sequence and return close to the target path with only moderate overshoot. In contrast, the uncontrolled vehicle exhibits a large path deviation after the second transition and fails to recover to the lane centerline, indicating loss of directional stability.

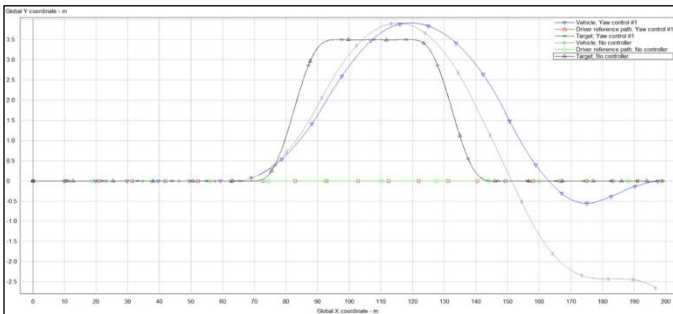


Figure 10: Reference Trajectory vs MPC tracking

This trend is also reflected in the steering-wheel [Fig 12] response. In the uncontrolled case, the driver model demands very large corrective steering inputs, with excursions approaching $\pm 500^\circ$, in an attempt to recover the vehicle motion. When the controller is active, the steering

demand is substantially smaller and better damped, which indicates that the direct yaw-moment action is effectively assisting the vehicle during the maneuver.

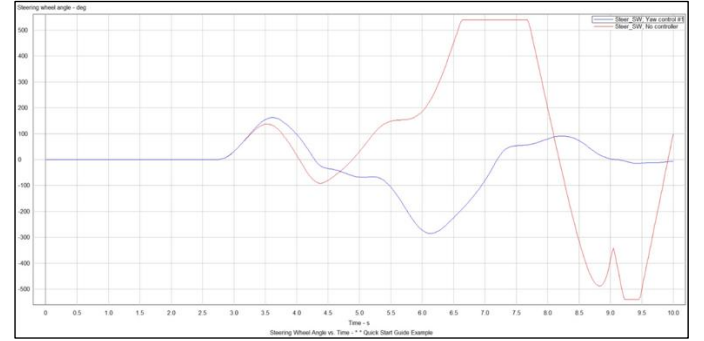


Figure 11: Steering Wheel Angle in the DLC maneuver

Results

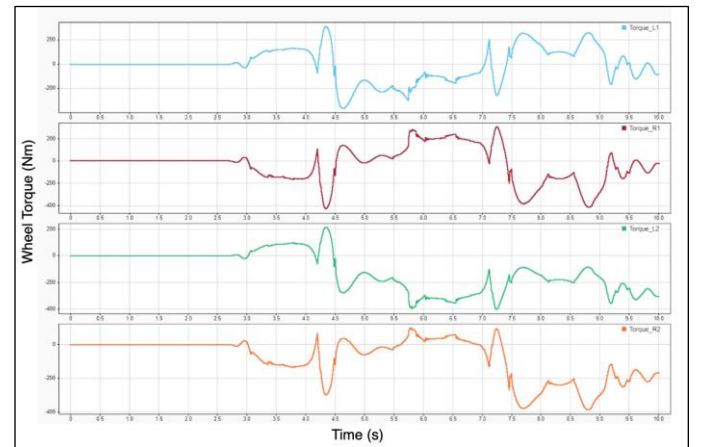


Figure 12: Wheel torque (1. Front left wheel 2. Front right wheel 3. Rear left wheel 4. Rear right wheel) vs time for low adhesion surface

The yaw-moment tracking [Fig 14] and wheel-torque [Fig 13] responses further explain the control mechanism. The actual yaw moment produced by the MPC allocator follows the desired yaw moment very closely over almost the entire maneuver, even during the most aggressive transients. This demonstrates that the lower-layer controller is able to realize the upper-layer yaw-moment demand with high fidelity despite the reduced-friction operating condition. The four wheel-torque histories show pronounced left-right and front-rear redistribution, together with several sign reversals during the two steering reversals. Such behavior is expected near the handling limit, where the controller must simultaneously generate corrective yaw moment and respect the available adhesion at each tire. Overall, the low-adhesion DLC results demonstrate that the proposed hierarchical fuzzy-MPC control strategy substantially improves path-following capability and prevents the loss of stability in a maneuver where the uncontrolled vehicle approaches the limits of handling.

The most significant improvement is observed in the yaw-rate [Fig 15] and sideslip responses [Fig 16]. Without control, the yaw rate becomes poorly damped after the first lane transition and later increases rapidly to approximately 70 degrees/s during the return phase, which is characteristic of an unstable response. At the same time, the sideslip angle departs from the nominal region and grows to a very large magnitude, eventually reaching approximately -140° , which indicates a near spin-out condition. By comparison, with the proposed controller the yaw rate remains bounded within a much smaller range, approximately -12 to 12 deg/s and the sideslip angle stays close to zero

throughout the maneuver with only minor transient deviations. These results confirm that the controller can preserve lateral stability even when the available tire-road adhesion is severely limited.

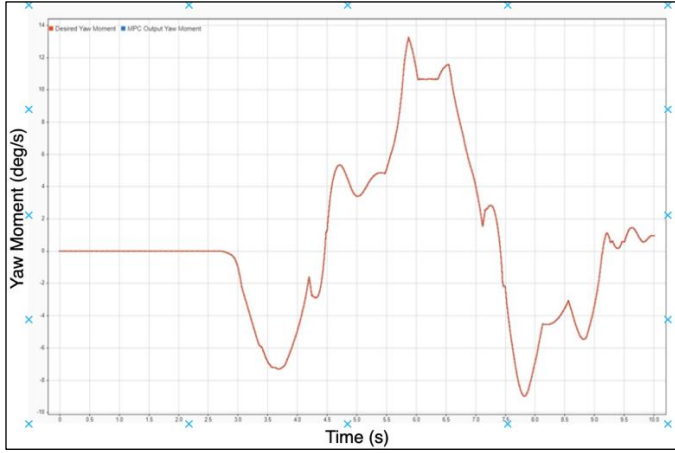


Figure 13: Yaw moment tracking for low μ DLC

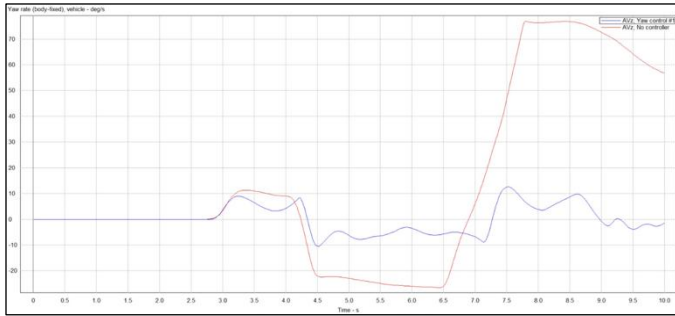


Figure 14: Yaw rate comparison (with MPC vs w/o controller)

The most significant improvement is observed in the yaw-rate [Fig 15] and sideslip responses [Fig 16]. Without control, the yaw rate becomes poorly damped after the first lane transition and later increases rapidly to approximately 70 degrees/s during the return phase, which is characteristic of an unstable response. At the same time, the sideslip angle departs from the nominal region and grows to a very large magnitude, eventually reaching approximately -140° , which indicates a near spin-out condition. By comparison, with the proposed controller the yaw rate remains bounded within a much smaller range, approximately -12 to 12 deg/s and the sideslip angle stays close to zero throughout the maneuver with only minor transient deviations. These results confirm that the controller can preserve lateral stability even when the available tire-road adhesion is severely limited.

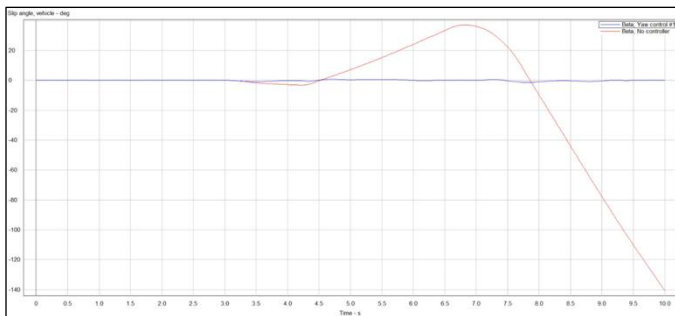


Figure 15: Side slip angle comparison (with MPC vs w/o controller)

C. Test Case 3: Sine Wave Steer Maneuver

The sine wave sweep maneuver excites the vehicle's yaw and lateral dynamics symmetrically in both directions over multiple cycles, which makes it more informative for assessing controller behavior, transient response, and steady-state error than the single-event dwell. The longitudinal speed was held constant at 100 km/h by the speed controller, and the maneuver was carried out on a dry asphalt surface with a friction coefficient of 0.85 over a simulation duration of 10 s. Two cases were compared: the baseline vehicle without any yaw controller, and the vehicle with the proposed MPC-based yaw-moment controller active.

The Y vs X plot [Fig 18] shows that the controlled vehicle follows a tighter, more contained trajectory, terminating near (210 m, 29 m), whereas the uncontrolled vehicle drifts laterally to approximately (220 m, 58 m) over the same period. The controlled trajectory is smoother and more symmetric about the intended path, consistent with the reduced yaw-rate and sideslip behavior.

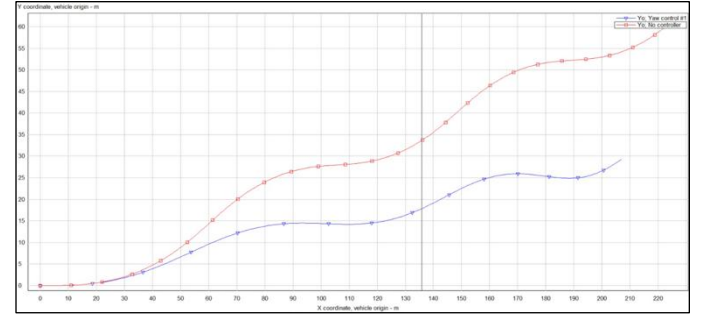


Figure 16: Trajectory Tracking for Sine wave steer input

Results

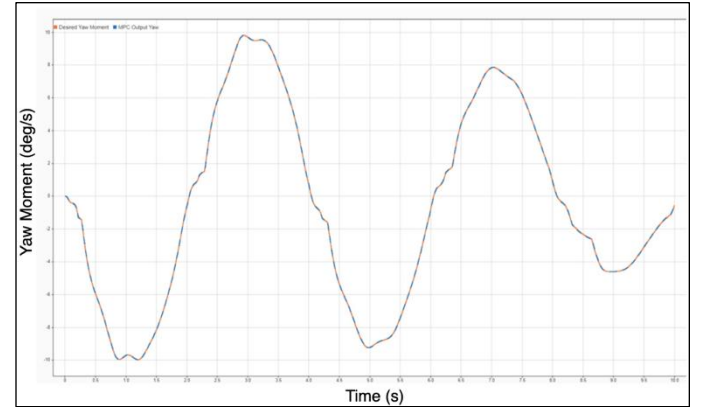


Figure 17: Yaw Moment tracking for Sine wave steer input

The MPC output yaw moment overlays the desired yaw moment computed by the reference model almost exactly across the full 10 s window, including at the local peaks near $t \approx 1.0$ s, 3.0 s, 5.0 s, and 7.0 s. The tracking error is negligible, confirming that the MPC is correctly solving the constrained optimization problem at every control step and

that the actuation allocation is delivering the commanded moment to the plant.

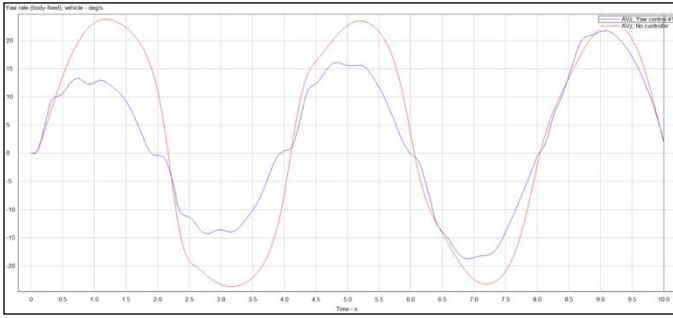


Figure 18: Yaw rate vs time (with MPC vs w/o controller)

Without the controller, the body-fixed yaw rate [Fig 20] reaches approximately ± 23 deg/s at the first and second cycles. With the controller engaged, the peak yaw rate is reduced to roughly ± 13 – 16 deg/s during the same cycles – a reduction of about 30–45 % at the peaks. The two responses converge near the end of the run as the steering amplitude and resulting demand fall within the linear handling region, indicating that the controller acts predominantly when the vehicle is approaching its non-linear limit, as intended.

The side slip [Fig 21] response shows the most pronounced improvement. The uncontrolled vehicle develops sideslip excursions of approximately $+5.7^\circ$ and -6.5° , which are well outside a stable handling envelope and indicate that the rear axle is losing lateral grip. With the MPC active, β is constrained within roughly $\pm 1.5^\circ$ throughout the maneuver – a reduction in peak sideslip of approximately 75 %. This is the key stability outcome: the controller is preventing the vehicle from departing into an oversteering condition.

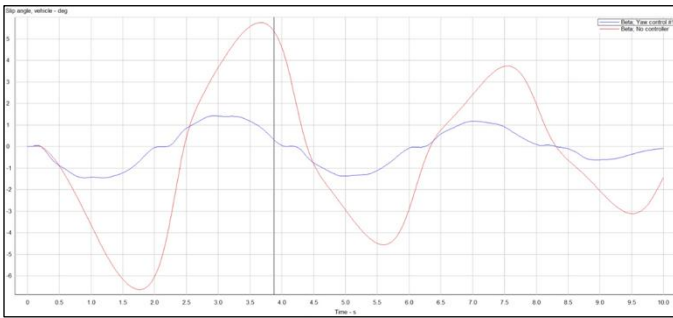


Figure 19: Side slip angle vs time (with MPC vs w/o controller)

Conclusion

This work proposed and validated a two-layer hierarchical stability control architecture for a four-wheel-independent-drive electric vehicle, in which a Mamdani-type fuzzy logic controller generates the corrective yaw moment from yaw-rate and sideslip-angle errors, and a constrained model-predictive controller allocates this demand among the four wheel motors by solving an online quadratic program with the Hildreth dual method. The simulation results demonstrate that the proposed strategy delivers consistent improvements in lateral stability across all three test cases. Taken together, these results confirm that the fuzzy-logic upper layer is effective at translating handling-state errors into a stabilizing yaw-moment demand, that the MPC lower layer is capable of realizing this demand in real time while explicitly respecting actuator and tyre-adhesion constraints, and that the two layers operate cooperatively without observable conflict. The controller intervenes predominantly when the vehicle approaches the

limits of handling and stays largely passive in the linear region, which is the intended behaviour for a stability augmentation system.

Accountability Statement

All three authors actively participated in the technical discussions related to controller formulation, tuning strategy, test case selection, and interpretation of simulation results.

Abhishek led the formulation and implementation of the MPC torque-allocation layer, including the slip-ratio-based prediction model, the constraint and cost-function design, and the Hildreth dual QP solver. Shantanu was primarily responsible for the CarSim plant model setup, configuration of the three test maneuvers, and the MPC tuning across all test cases. The integration of the Simulink and CarSim co-simulation framework, along with the 2-DOF reference model and fuzzy upper-layer refinement, has been carried out jointly by Srujan and Shantanu. In addition, all authors have been actively involved in debugging, troubleshooting, and refining the simulation model.

The report writing was led primarily by Srujan, while Abhishek contributed to the document layout and formatting. The presentation was prepared collaboratively by all three authors, with Shantanu taking the lead in organizing and developing the presentation flow. All authors reviewed the final manuscript and presentation material, and all accept shared responsibility for the technical accuracy, integrity, and overall quality of the work.

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Appendix

Vehicle parameter	Symbol	Value	Unit
Total vehicle mass	m	1724.7	kg
Sprung mass	m_s	1430	kg
Wheelbase	L	2.66	m
CG to front axle	a	1.05	m
CG to rear axle	b	1.61	m
Track width (f, r)	t	1.49	m
Height of CG	h	0.57	m
Yaw inertia	I_z	2723	kg·m ²
Front cornering stiffness	C_f	183350	N/rad
Rear cornering stiffness	C_r	114590	N/rad
Tyre size	—	265/65 R17	—
Wheel rolling radius	R_w	0.31	m
Steering ratio	i_s	20:1	—
Max motor torque	T_{max}	±500	Nm
Stability factor	K	-0.0053	rad